

Tutorial-1 (Solution)

Ans: to the question numbers 1

(a-i) Digital Image: A digital image is a 2D representation of visual scene where the image is composed of a finite set of discrete elements called pixels. Each pixel has a specific location and an associated value.

(a-ii) Bit plane: A bit plane represents the image data corresponding to a particular bit position in the binary representation of pixel values.

(a-iii) Image dithering: Image dithering is the process used to simulate colors or shades that are not available in a limited color palette.

(2)

(Q-iv) Lookup Table: A lookup table is a predefined table that maps input pixel values to new output values for image transformation.

b) The types/file formats of digital images:

- 1) BMP (Bitmap) → Uncompressed; Large file size; high quality
- 2) JPEG/JPG (Joint photographic Experts Group): Lossy Compression; used for photos
- 3) PNG (portable Network Graphics) → Lossless Compression; supports transparency -
- 4) TIFF/TIF (Tagged Image File Format) → High quality; often used in printing
- 5) GIF (Graphic Interchange Format) → Support animation, 256-color

c-1) Application of Digital Image processing:-

- 1) Medical Imaging:- MRI, CT scan, X-ray enhancement, tumor detection
- 2) Remote sensing/satellite Imaging:- Land-use mapping, forest monitoring, weather prediction
- 3) Computer Vision:- Face recognition, object detection, autonomous vehicles
- 4) photography/media:- Image enhancement, filtering, color correction, noise removal
- 5) Security and Surveillance:- license plate recognition, facial authentication, motion detection
- 6) Industrial Inspection:- Detecting defects in products, quality control
- 7) Biometrics:- Fingerprint, Iris, facial pattern recognition
- 8) Document processing:- OCR, handwriting recognition

c-i) Different Energy Source used to create a digital Image

- 1) Visible Light :- Normal photography, Video Cameras
- 2) Infrared Radiation :- Night vision, heat maps, remote Sensing
- 3) X-ray :- medical Imaging, Air-port security scanner
- 4) Ultra sound :- Medical ultrasound (body scan)
- 5) Magnetic field :- MRI
- 6) Microwave :- Weather radar, Satellite communication
- 7) Radar :- Remote sensing, aircraft surveillance
- 8) Electrons :- Electron microscopy

Ans: to the Q: N: 2

a) Image Histogram :- An image histogram is a graphical representation that shows how the pixel intensities are distributed in an image.

- ⊗ The x-axis $\rightarrow$ pixel intensity values (0-255)
- ⊗ The y-axis $\rightarrow$ frequency of pixel

Why: H.E is a technique used to enhance the Contrast of an image

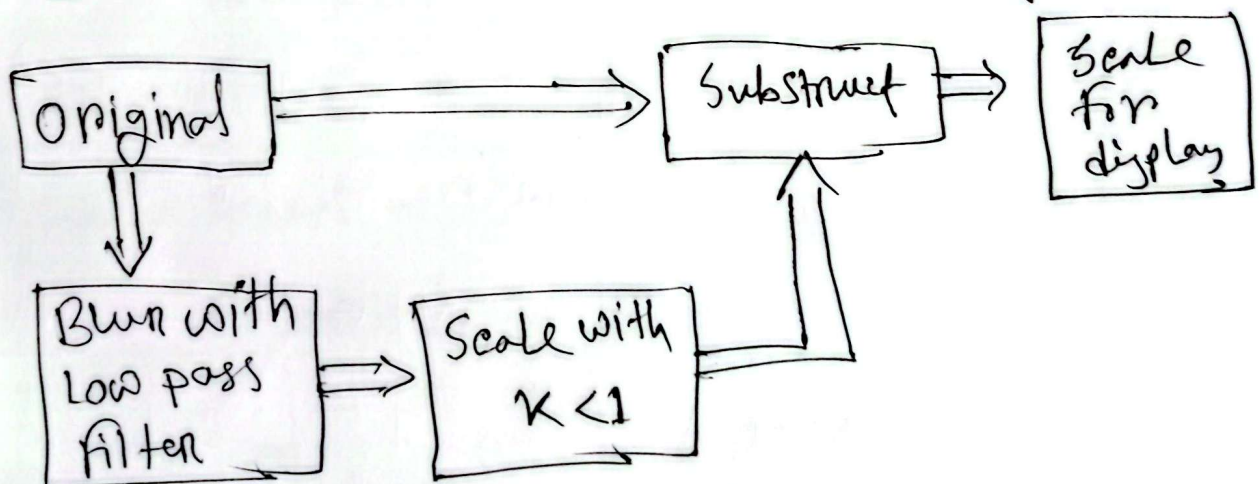
When:

- 1) If the image has Low contrast
- 2) Detailed are hidden in shadow
- 3) To enhance visibility

b) Component of grayscale image: Component of grayscale image refers to the subtraction of each pixel value from light gray value for 8-bit image (I),

$$I' = 255 - I$$

c-i) The schema for Unsharp Masking:



C-ii) PGM $\rightarrow$ portable GrayMap file-format

Structure of a PGM file:-

1) Magic Number:

⊗ P2 $\rightarrow$ ASCII (text format)

⊗ P5 $\rightarrow$ Binary (raw format)

2) Comment Line:

Begin with #; used for description

3) Image Dimensions: no. of row and column

4) Maximum Gray value: Typically 255 (for 8 bit)

C-in) Filtering process with a mask on an image

Filtering is the process of modifying pixel values in an image using a filter mask that slides over the image

Steps

1) Select a filter mask (3x3 matrix)

Example:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

④

- 2.) place the mask center over each pixel in the image.
- 3.) Multiply corresponding pixel and mask values
- 4.) ~~sum~~ the result $\rightarrow$ replace center pixel with it.
- 5.) Move the mask to the pixel and repeat until the entire image is processed

Mathematical Representation :-

$$g(x,y) = \sum_i \sum_j f(x+i, y+j) \cdot w(i,j)$$

Where;

$f(x,y)$: input image

$w(i,j)$: filter mask

$g(x,y)$: output image

Ans: to the Q: N: 3

Q) ROI:- ROI filtering means applying image processing or filtering operations only to a specific selected region of an image instead of the entire image

b)

Feature	RGB Image	Indexed Color Image
Definition	Each pixel store actual Red, Green, Blue intensity value	Each pixel store an index that refer to a color in a color table
Storage	Require more memory	Require less memory
Representation	3D matrix	2D matrix
Color Range	up to 16.7 million colors	Limited by palette size
Use case	photography, high-quality image, real-world scene	Gifs, icons, diagram and graphics with limited colors
Example	.jpg, .png	.gif, .bmp

c) Separable Filter :- A Separable filter is a 2D filter that can be broken down into two 1D filter - one for rows and one for columns. This reduce computational cost from N^2 to N .
 If a 2D filter mask $H(x, y)$ can be expressed as $H(x, y) = h_1(x) \cdot h_2(y)$

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Example :

$$\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \begin{bmatrix} 1, -2, 1 \end{bmatrix}$$

Ans: to the Question N: 4

a) Expression for 2D Gaussian filter

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

b) Importance of Gaussian filter

1) Noise reduction: Effectively removes high-frequency noise

2) Smooth transitions: Unlike box filters, Gaussian filters produce smooth, natural blurring without harsh edges

3) Isotropic behaviour: Applies equally in all directions

4) Separable properties: Can be implemented using two 1D convolution

5) Used in edge detection: preprocessing step in algorithms like Canny Edge Detection to remove noise

6) We obtain an averaging filtering as a limiting case of a Gaussian filter when the standard deviation ($\sigma \rightarrow \infty$) become very large -

Ans: to the Q: N: 5

Feature purpose	Avg. Filtering	Gaussian Filtering	High-pass Filtering
	Smooths the image with weighted average neighbour pixel value	Smooth Smooths the image with weighted average	Enhance edges and details by removing low frequencies

(11)

Feature	Avg. filtering	Gau. filtering	High-pass filtering
Type	Low-pass	Low-pass	High-pass
Weight distribution	All pixel have equal weights	Center pixel have higher weight	Weights sum to zero
Effect on image	Remove sharp changes and noise	produce smooth natural blurr	Highlight edge, lines and fine detailed
Used in	Noise reduction	preprocessing for edge detection	Edge detection

b-i) Approach to deal with border problem in filter

- 1) Zero padding: Assume pixels outside the image are zero
- 2) Replication padding: Extend the nearest border pixel values outward.
- 3) Reflection: Mirror the border pixels across the boundaries
- 4) Circular: Wrap the image around as if it repeats cyclically.

b-ii)

- 1) Image Acquisition: Capture the envelope image using a camera
- 2) preprocessing: Convert to grayscale, remove noise, enhance contrast
- 3) Segmentation: separate text region from background
- 4) Character Recognition: - Use OCR to detect digit
- 5) Postcode Extraction: locate and extract the region

b-iii) Broad Subclasses of Image processing Algo

- 1) Low-level processing: Basic pixel-based operations
 - 1) Noise removal, contrast stretching, blurring
- 2) Mid-level processing: Extract features and object
 - 1) Segmentation, edge detection
- 3) High-level processing: Interprets image content
 - 1) Object classification, scene understanding, decision making

Tutorial-2 (Solution)

Ans: to the Question Number 1

a-i) Image Moment :- Image moment are quantitative measures used to describe the shape and intensity distribution of an image.

a-ii) Color Channel :- A color channel represents one components of a color model used to form a full-color image.

a-iii) Image Segmentation :- Image Segmentation is the process of dividing an image into meaningful regions or objects. It separates foreground and/or from background.

a-iv) Entropy :- measures the amount of information or uncertainty in an image. It is derived from information theory.

b) Four Common noise in digital images:-

1) Gaussian Noise: Random variation in intensity following a normal distribution.

2) Salt and Pepper Noise: Random black and white pixel appear in the image

3) Speckle Noise: Multiplication noise appearing as granular-spots.

4) Poisson Noise: Noise that depends on Image intensity - brighter area has more noise

c) 3-bit gray code (0 to 7)

Decimal	Binary	3-bit gray
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100

Ans! to the Question Number 2

a) Classification of image segmentation technique:-

Image segmentation technique is the process of dividing an image into meaningful regions -

Segmentation technique are broadly classified into four main categories:-

1) Threshold-based methods:-

Global/local thresholding → separate region based on the pixel intensity -

2) Edge-based methods: Edge detection (

Sobel, Canny) → detect object boundaries where there is a sharp change in intensity -

3) Region-based methods: Region growing, splitting, merging → Group neighboring pixels with similar properties

4) Clustering-based methods: K-means, Fuzzy c-means → Classify pixel into groups based on similarity measurement

5) Model-based methods: Active contours, watershed -

b) Difference between Lossless and Lossy image compression:-

Aspect	Lossless	Lossy
Definition	Compresses data without losing any information. The original image can be perfectly restored	Some data is permanently lost during compression. Image can't be fully recovered -

Quality	No degradation in Quality	Slight Loss in quality
Compression Ratio	Lower (2:1 or 3:1)	Higher (up to 100:1)
Example	PNG, GIF, TIFF	JPEG, MP3
Use Case	Medical Images,	photograph

C-i) HSV color model

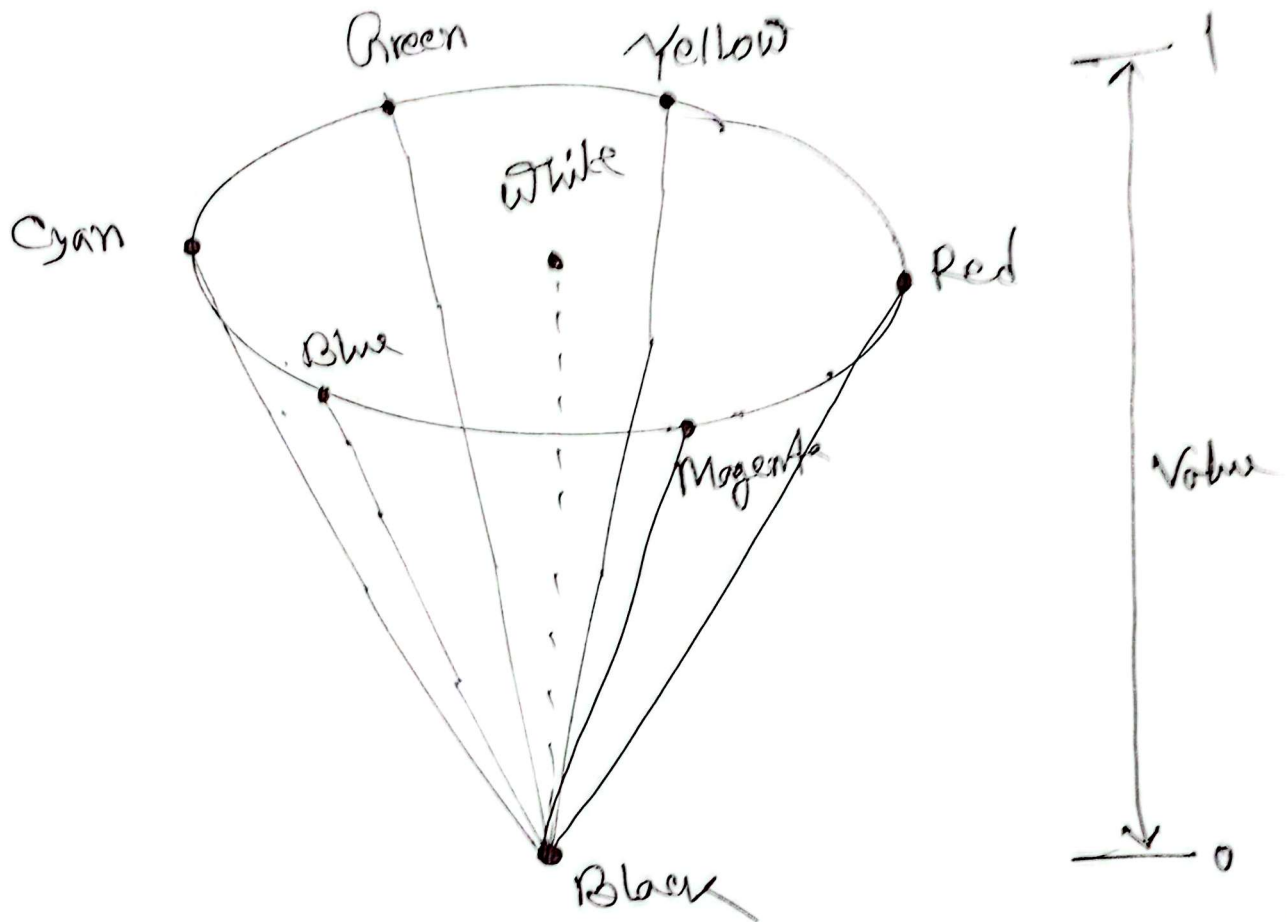
HSV stands for hue, saturation, and value. These terms have the following meaning:

Hue: The color attributes (red, green, blue, yellow, and so on)

Saturation: The amount by which the color has been diluted with white. The more white in the color, the lower the Saturation.

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Value:- The degree of brightness: a well-lit color has high intensity, a dark color has low intensity.



c-ii) Image restoration Model:-

Image restoration is the process of recovering the original image from a degraded or blurred version by removing noise, blur, or distortion that occurred during image acquisition, transmission or storage.

The basic model

The degradation model of an image is written as:

$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$$

here;

$f(x,y)$ = original image

$h(x,y)$ = Degradation function

$\eta(x,y)$ = Additive noise

$g(x,y)$ = Observed degraded image

$*$ = Convolution Operator

To estimate $f(x,y)$ (the original image) from the observed $g(x,y)$, given knowledge or assumption of $h(x,y)$ and $\eta(x,y)$

that is

$\hat{f}(x,y)$ = Restoration process ($g(x,y)$)

$$\hat{f} = \frac{g(x,y) - \eta(x,y)}{h(x,y)}$$

Ans: to the Question number 3

a) CMYK model

The CMYK color model is a subtractive color model used mainly in color printing. It represents colors as a combination of four inks:

Cyan (C), Magenta (M), Yellow (Y), Black (K)

Formula

$$C = 1 - R, \quad M = 1 - G, \quad Y = 1 - B$$

$$K = \min(C, M, Y)$$

b(i)

Aspect	Single threshold	Double threshold
No. of threshold	uses one threshold value	uses two threshold values (T_1, T_2)
Condition for pixel classification	pixel become white if its gray level $> T$ otherwise black	pixel become white if its gray level lies between (T_1, T_2) otherwise black

purpose	Separates objects from background based on a single cut-off	Isolate objects within a specific range of intensity-
Output	Binary image with broad segmentation	Binary image focusing on features in a narrower intensity band
Use case	Simple object detection, when foreground and background are simply separable	Edge/boundary detect highlighting structures

b-ii)

Feature	Additive noise	Multiplicative noise
Definition	Noise that is added directly to the image	Noise that multiplies the image signal
Mathematical Model	$g(x,y) = f(x,y) + n(x,y)$	$g(x,y) = f(x,y) * n(x,y)$
Dependency on image	Independent of the image content	Dependent on the image pixel intensity
Example of noise type	Gaussian noise, Uniform noise	Speckle noise, Rayleigh noise-

c) Where thresholding is applied :- Thresholding is widely applied in image segmentation tasks where we need to extract meaningful objects or regions from the background - ⁽¹²⁾

Application of thresholding :-

- 1) Document Image Analysis :- Separating text or handwriting from the paper background
- 2) Medical Image processing :- Extracting tumors, bones or tissues from MRI, CT, or X-ray scans.
- 3) Industrial Inspection :- Detecting defects, cracks, or flaws in materials or manufacturing components.
- 4) Object detection :- Identifying simple shapes or objects in a scene based on intensity difference
- 5) License plate Recognition :- Isolating license plate regions from vehicle image for character.
- 6) Face Detection and Biometric :- Detecting regions of interest such as faces, fingerprints

Ans: 4 the Q: N: 4

a) Method-1 each line encoded separately by the count of first 0's then 1's then 0's and so on — so the encoded version is
 $(123) (231) (0,321) (141) (0132)$

Method-2 in this method, each row encoded as the position and count of 1's in the row. So the encoded version is:

$$(22) (33) (361) (24) (4,3) (1152)$$

$$b) V = \max(0.2, 0.4, 0.6) = 0.6$$

$$\delta = V - \min(0.2, 0.4, 0.6) = 0.4$$

$$\zeta = \frac{0.4}{0.6} = 0.667$$

Since $B = \zeta$ we have

$$A = \frac{1}{6} \left(4 + \frac{0.2 - 0.4}{0.4} \right) = 0.5833 \quad \text{Am}$$

Ans: to the Q: N: 5

```

a) import cv2
import matplotlib.pyplot as plt

img = cv2.imread(img-path)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

plt.imshow(img_rgb)
plt.title("RGB Color Image")
plt.axis('off')
plt.show()

```

b-i) Huffman Coding Steps

- 1) Determine the probabilities of each gray value in the image
- 2) form a binary tree by adding probabilities two at a time, always taking the two lowest available value
- 3) Now assign '0' and '1' arbitrarily to each branch of the tree from its apex
- 4) Read the code from the top

b+ii) Moments are statistical measures of an image that describe its shape characteristics

Definition:—

⊗ Image moments are derived from pixel intensity, used to calculate centroid, area, Orientation, etc.

⊗ Moment invariants are combinations of moments that remain unchanged under:

- 1) Translation
- 2) Rotation
- 3) Scaling

Hu's Seven Moment Invariants:—

Introduced by Hu(1962), These are computed from normalized central moments and are used in pattern recognition

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b (iii) Edge Detecting Techniques/mask

1) Sobel Operator :- Use 3×3 matrix mask

Detects horizontal and vertical edges

Smooths noise slightly

2) Prewitt Operator :- Similar to Sobel -

Simpler approximation of gradient.

Less smooth than Sobel

3) Robert Cross :- use 2×2 mask $\rightarrow$ Detects

diagonal edge; direction sensitive to noise

4) Laplacian Operator :- Used second derivative

Detects all edge directions at once;

Very sensitive to noise

5) Canny Edge Detector :- Multi-stage (gradient, threshold, hysteresis) most accurate

and robust edge detection method -

Tutorial-3 (solution)

Ans: to the Question Number 1

- i) Image Topology:- Image topology refers to the spatial relationships and connectivity between pixels or regions in a digital image
- ii) Chain code:- A Chain code represents the boundary of an object as a sequence of directional moves between connected boundary pixels.
- iii) Shape Descriptors:- Shape descriptors are numerical features that describe the geometry or outline of an object in an image.
- iv) FFT:- FFT is a fast computational algorithm to calculate the discrete Fourier Transform of an image or signal

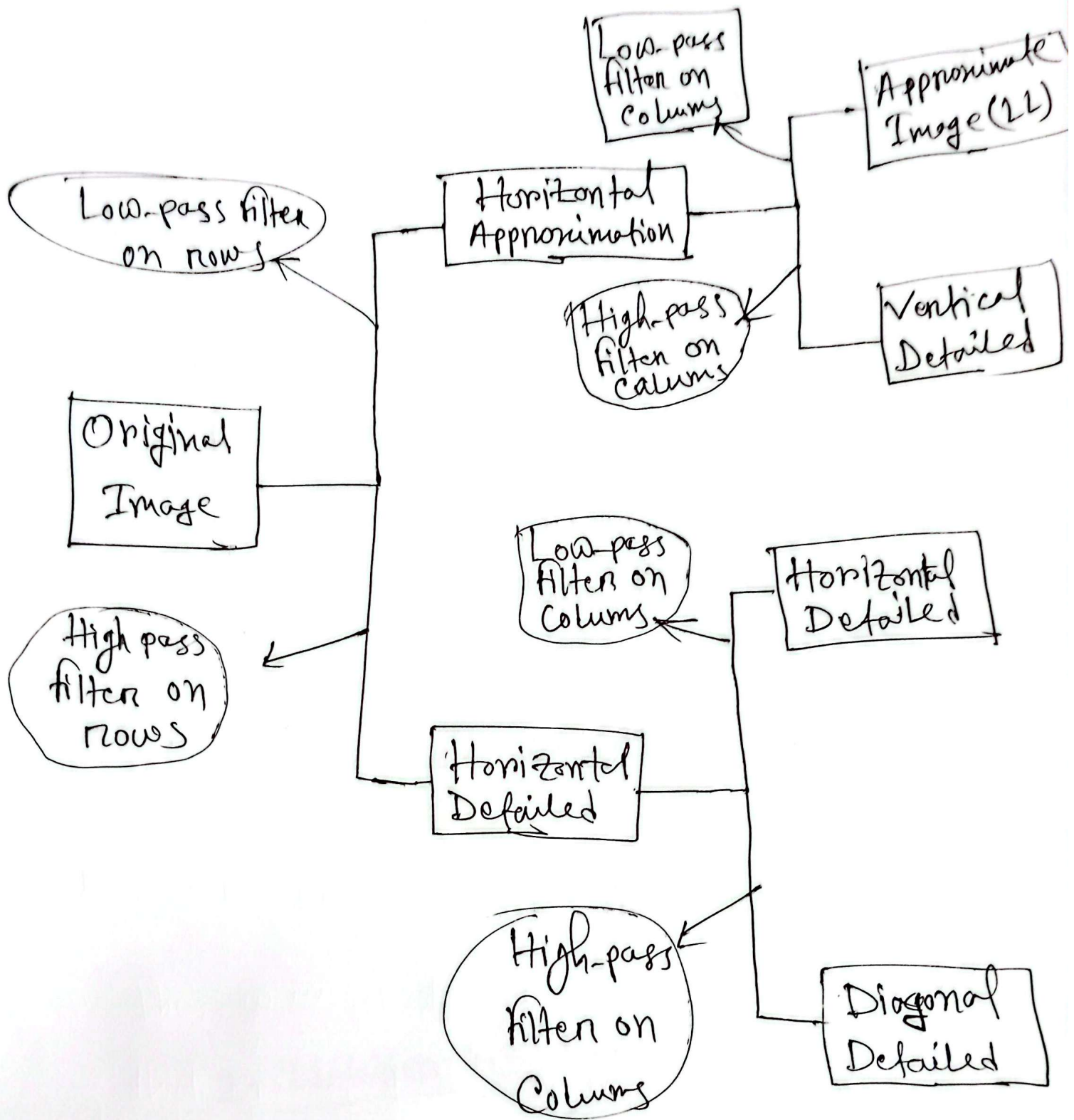
v) Boundary of Objects:- The boundary of an object is the set of pixels that separate the objects from the background in a digital image

vi) Special Effect:- In image processing, special effects refer to artistic or visual transformation applied to images to enhance or modify their appearance

Ans! to the Q! N! 2

i) Basic principle of wavelet:-

The wavelet transform is a mathematical tool that breaks down an image into different frequency components at various resolution



ii) Morphological Operation on image:-

Morphological image processing deals with shape-based operations on binary or grayscale image using a small matrix called a structuring element.

Basic operation:-

1. Erosion ($\ominus$) :-

⊗ Shrinks or thins objects

⊗ Removes small white noise and detach connected component

Formula:

$$A \ominus B = \{x | B_x \subseteq A\}$$

2. Dilation ($\oplus$) :-

⊗ Expands objects or fills small holes

⊗ Adds pixels to the boundaries of objects

* Formula

$$A \oplus B = \{2 \setminus (B) \cup \cap A\}$$

3. Opening (o):-

* Erosion followed by dilation

⊗ $(A \ominus B) \oplus B$

⊗ Remove small objects (noise) while keeping the main shapes

4. Closing (.):-

⊗ Dilation followed by erosion

⊗ $(A \oplus B) \ominus B$

⊗ Fills small holes or gaps inside the objects -

Ans: to the Q: IV: 3

a)

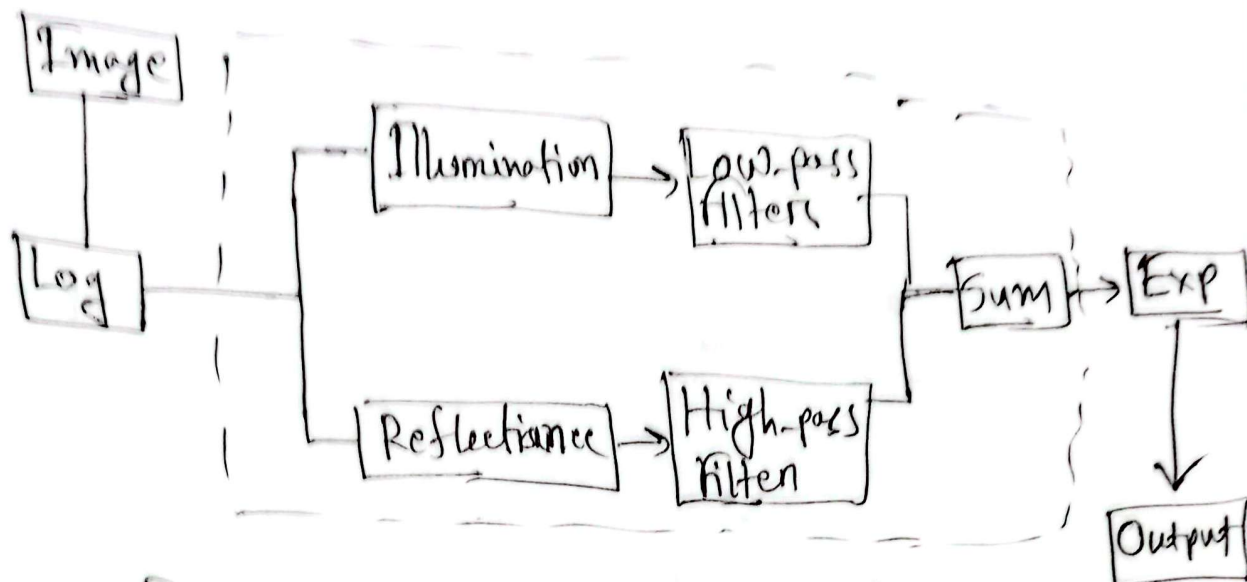


Fig: A Schema for homomorphic filtering

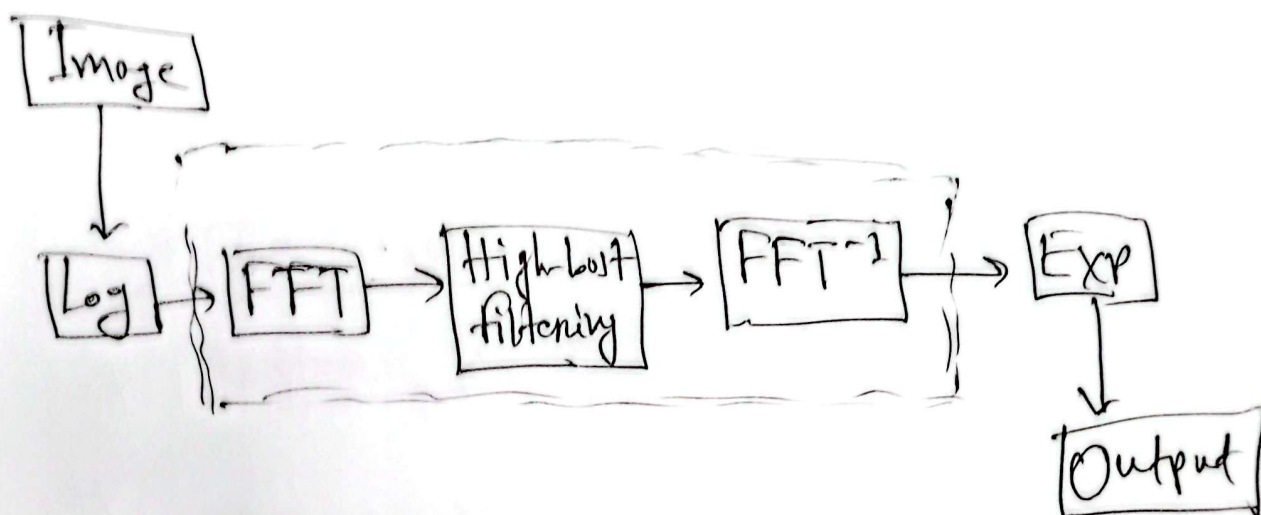


Figure: A simple Schema for homomorphic filtering

b) Application of special effects:-

- 1) Film and Entertainment Industries
- 2) Advertising and Graphics Design
- 3) Augmented and Virtual Reality
- 4) Medical Imaging
- 5) Photography and Art creation
- 6) Education and Research

Ans! to the Q: No. 9

a) Four most common used wavelet:-

- 1) Haar Wavelet:- The simplest and oldest wavelet
- 2) Daubechies Wavelets:- provide better smoothness and frequency localization
- 3) Symlet Wavelet:- A modified and more symmetrical version of Daubechies
- 4) Coiflet Wavelets:- Have better orthogonality and vanishing moments-

b) Let $f(n)$ be a discrete function with only four values

$f(0)=2$, $f(1)=3$, $f(2)=2$, $f(3)=1$ the size of L is 4 hence $L=4$

$$F(0) = \frac{1}{L} \sum_{n=0}^3 f(n) e^{i2\pi n \cdot 0} = \frac{1}{L} \sum_{n=0}^3 f(n) e^0$$

$$= \frac{1}{4} \sum_{n=0}^3 f(n)$$

$$= \frac{1}{4} (f(0) + f(1) + f(2) + f(3))$$

$$= \frac{1}{4} (2+3+2+1) = \frac{1}{4} \times 8 = 2$$

$$F(1) = \frac{1}{4} \sum_{n=0}^3 f(n) \left[\cos \frac{2\pi n}{4} + i \sin \frac{2\pi n}{4} \right]$$

$$= \frac{1}{4} (2(1+i) + 3(0+1) + 2(-1+i) + 1(0-i))$$

$$= \frac{1}{4} (2+2i-2+i) = \frac{1}{4} \cdot 3i = \frac{3}{4}i$$

$$F(2) = \frac{1}{4} \sum_{n=0}^3 f(n) \left[\cos \frac{4\pi n}{4} + i \sin \frac{4\pi n}{4} \right]$$

$$= \frac{1}{4} [2(1+0) + 3(-1+0) + 2(1+0) + 1(-1+0)]$$

$$= \frac{1}{4} (2-3+2-1) = 0$$

$$F(3) = \frac{1}{4} \sum_{n=0}^3 f(n) \cdot \left[\cos \frac{2\pi n}{4} + i \sin \frac{2\pi n}{4} \right]$$

$$= \frac{1}{4} [2(1+0) + 3(0-i) + 2(-1+0) + i(0+i)]$$

$$= \frac{1}{4} (2 - 3i - 2 + i)$$

$$= \frac{1}{4} - 2i = -i/2$$

$$\left\{ 2, \frac{1}{2}, 0, -\frac{1}{2} \right\}$$

Ans: to the Q: N: 5

a-i) import cv2

import numpy as np

import matplotlib.pyplot as plt

img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

dft = cv2.dft(np.float32(img))

dft_shift = np.fft.fftshift(dft)

magnitude = cv2.magnitude(dft_shift[:, :, 0],
dft_shift[:, :, 1])

Magnitude: `spectrum = np.log(1 + magnitude)`

`plt.figure(figsize=(10,5))`

`plt.subplot(1,2,1)`

`plt.imshow(img, cmap='gray')`

`plt.title('Original grayscale Image')`

`plt.axis('off')`

`plt.subplot(1,2,2)`

`plt.imshow(magnitude_spectrum, cmap='gray')`

`plt.title('DFT magnitude Spectrum')`

`plt.axis('off')`

`plt.tight_layout()`

`plt.show()`

Explanation:-

- 1) Read grayscale image using OpenCV
- 2) `cv2.dft()` convert the image to frequency domain
- 3) `np.fft.fftshift()` centers the low-frequency component

- 3X
- 4) The magnitude spectrum ($\log(I+ \text{magnitude})$) makes frequency detailed visible
 - 5) Display both original and frequency domain version using matplotlib.

a-ii)

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread(img_path, cv2.COLOR_BGR2RGB)
```

threshold = 128

Solarized = np.where(img < threshold, img, 255 - img)

```
plt.figure(figsize=(10,5))
```

```
plt.subplot(1,2,1)
```

```
plt.imshow(img)
```

```
plt.title('Original image')
```

```
plt.axis('off')
```

```
plt.subplot(1,2,2)
```

```
plt.imshow(Solarized)
```

```
plt.title('Solarized image')
```

```
plt.axis('off')
```

```
plt.show()
```

Explanation:-

- 1) Load color image and convert to RGB for display
- 2) If pixel intensity > 128 , it is inverted (255-value) else left unchanged-
- 3) This creates a psychedelic/negative-like effect known as Solarization
- 4) Display both image side by side

b)

Aspects	Mask	Structural Element
Purpose	Used for linear or non linear filtering	Used to analyze or modify shapes in binary or grayscale images
Operation Type	Works on intensity values	Works on objects shape and boundary
Example	Average filtering, Sobel filter, Gaussian	Dilation, Erosion, Opening, closing
Mathematical Basis	Convolution or Correlation	Set theory/Logical operation
Nature	Numeric weight	Binary pattern